

Event-driven

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The event-driven architecture is based on the concept of a stream of data. Each stream of data raises an event. What is event used for? It can be used for services or components to communicate with each other.

- **Producer** is the service/application that produces events
- The event is communicated and routed to the receivers. This is called the **broker**. It is the middleware between producer and consumer
- Consumer is the service/application that consumes events

There are some patterns for event-driven

- We can store the events in an event source. The events can be stored in the order they arrive. This pattern is known as the Event Sourcing Pattern.
- The key benefit of this pattern is that it allows us to replay the events. This could be useful in a system where we want to skip to the step that failed and retry it post-fixes
- We can also have separate services where one service can subscribe to the read event service and another service subscribes to the write events. This pattern is known as CQRS

Event bus/pub-sub/observer pattern

Can read here for more detail <https://medium.com/frontend-canteen/publish-subscribe-pattern-in-javascript-17bf1e94e83d>

They all pretty much the same.

the publish-subscribe pattern is a messaging pattern.

A pub-sub is like a panda having a loudspeaker to announce important news to all the other pandas in the enclosure. For example, the panda might warn his panda friends about a predator. This happens in real-time. And there is no guarantee that everyone will get the message. All the pandas that can hear an announcement can choose to act on it or ignore it.

How to implement pub-sub pattern in Rabbit MQ: <https://write.agrevolution.in/rabbitmq-pubsub-exchange-and-queue-management-67e5a465f59b>

Message queue

Provide asynchronous communication protocol, the producer and consumer don't need to interact at the same time.

In this model, messages are sent from a sender to a receiver via an intermediate message queue, which stores messages until they can be processed by the receiver. Message queues are typically used to decouple components and provide fault tolerance by allowing messages to be retried in the event of a failure

Email Inbox: Think of your email inbox as a message queue. Emails are sent to your inbox asynchronously, and you can choose to read, reply to, or delete each email individually

What is message broker?

What is the difference between SNS and SQS?

<https://medium.com/awesome-cloud/aws-difference-between-sqs-and-sns-61a397bf76c>